Number 7 — Recurring decimals · Advanced calculator problems

Question 1 and 4 without a calculator. Marks in brackets. ★ = harder question. Dot notation: $0.\dot{2}\dot{7} = 0.272727\ldots$

1 Convert each recurring decimal to a fraction in its simplest form. [12]

a $0.\dot{7}$ **b** $0.\dot{2}\dot{7}$ **c** $0.4\dot{2}3\dot{}$ **d** $0.04\dot{5}$ **e** $0.8\dot{3}$ **f** $2.1\dot{6}$

2 Calculate, giving each answer correct to 3 significant figures. [6]

a $\dfrac{1}{2^3} + \dfrac{1}{3^3} + \dfrac{1}{4^3}$ **b** $\dfrac{6.4 \times 10^4}{8.15 \times 10^6}$ **c** $\sqrt[5]{\dfrac{7.2 \times 10^4}{3.1 \times 10^{-2}}}$

3 Find the value of z to 3 significant figures, given that $z = \sqrt[3]{\dfrac{a^2 + b^2 + c^2}{\tan(b-c)^\circ}}$ where $a = 40$, $b = 50$ and $c = 20$. [4]

4 a and b are digits. Show that $1.\dot{a}\dot{b} = \dfrac{ab + 99}{99}$, where ab is the two-digit number with digits a and b. [3]

Total: 25 marks

Check: one repeating digit → ×10, two → ×100, three → ×1000, then subtract · a non-repeating digit after the point: multiply by 10 first, then by 10 again (×100) and subtract · always simplify the fraction · calculator: brackets round numerators and denominators, keep full accuracy, round at the end.